11 Computer Vision and Artificial Intelligence for Intelligence Automation Systems (IAS)

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Abstract

Intelligence automation systems (IAS), which make use of computer vision and artificial intelligence (AI), are changing the way businesses operate by simplifying tasks reducing errors, and providing insights. These systems are necessary in sectors including banking, insurance, healthcare, and retail to ensure security and maintain quality control. To analyze data, find trends, and support decision-making processes, they rely on AI systems. We may expect improvements in automation systems that will become essential for organizations as technology develops. These advancements will be fuelled by emerging technologies like machine learning, edge computing, and IoT to further enhance AI vision solutions' capabilities. In security and surveillance applications, computer vision is taking over human monitoring duties by analyzing real-life scenarios to identify threats and deliver instant security assessments. Computer vision is transforming quality assurance techniques and production processes in manufacturing and automation settings to increase flexibility and efficiency. Additionally, it is used in agriculture to identify diseases and weeds, and it plays a critical role in the automobile sector by identifying quality flaws prior to the release of vehicles from the assembly line. AI and computer vision integration into business operations is changing how companies operate and opening doors for innovations that will increase efficiency and competitiveness in the marketplace.

Keywords: Computer vision, artificial intelligence (AI), automation, intelligence, machine learning, natural language processing (NLP)

11.1 Introduction

The rise of automation systems powered by computer vision (CV) and artificial intelligence (AI) [1] is causing a shift in how businesses operate across various industries. These advanced systems aim to streamline tasks boost efficiency reduce errors and offer insights into business operations [2]. CV, a subset of AI, focuses on enabling machines to interpret and comprehend data from their surroundings. This technology is crucial

in automation setups as it analyzes images and videos meticulously to identify patterns and make decisions based on the collected information. As a result, there have been enhancements in the speed, precision, and quality of business processes giving organizations the advantage needed to thrive in today's dynamic business landscape [3].

When used in automation frameworks for quality control procedures, CV technology offers benefits. Through the use of algorithms to identify flaws in product photos, businesses can quickly resolve problems and enhance the quality of their products while cutting down on waste and saving money and effort. Furthermore, CV is useful in surveillance and security applications. These systems scan video feeds and promptly notify security personnel of potential threats, such as trespassers or suspicious activity. This is particularly useful in high-risk areas like airports, government buildings, and military facilities.

Not only is the way organizations operate changing as a result of the integration of CV and AI but it is also opening the door for innovations that will increase productivity and market competitiveness [4]. AI systems examine data to look for trends and support the use of such data in decision-making. This technology is especially useful in situations where there is a lot of data to analyze, like the banking and insurance industries, where large volumes of financial data are examined by AI algorithms to look for trends that can point to fraud.

Another use of AI in automation systems is seen in healthcare, where through examining information AI algorithms can detect patterns that might indicate the presence of a disease or ailment [5]. This technology proves valuable in early disease identification like cancer, where timely intervention can greatly enhance outcomes. Smart automation systems incorporating CV and AI are also applied to enhance customer service. By examining customer information, these systems can recognize trends in customer behaviors and preferences enabling businesses to offer marketing strategies and customer service.

Automation systems that evaluate consumer data and provide shoppers with tailored recommendations have revolutionized the retail sector. Customers' purchasing experiences are enhanced by these systems.

Encourage sales by making recommendations for goods that fit their tastes. These systems are able to determine the preferences of their users by looking at their past purchases and online activity. Offer something that you know will grab their attention. Sales are not increased by this. Additionally, it increases brand confidence and client loyalty. As a result, intelligent automation solutions are now a benefit for shops hoping to prosper in the current business environment.

In today's world, advanced automation systems utilizing CV and AI are reshaping how businesses function [6]. By streamlining tasks, enhancing efficiency, and offering insights into business processes, these systems empower businesses to stay ahead in a rapidly evolving and competitive environment. The continuous evolution of cutting-edge technologies like machine learning, edge computing, and the Internet of Things (IoT) [7] indicates a future for automation systems. These progressions will lead to adaptable and scalable AI vision solutions that will revolutionize business operations and customer interactions.

11.2 Artificial Intelligence

AI [8–11] is an extensive field that encompasses the development of computer systems capable of exhibiting intelligent behavior [12]. These systems are made to use a variety of sensors and data sources to sense their surroundings, learn from prior interactions and experiences, and use this knowledge to accomplish predetermined objectives [13].

AI technology is now a vital part of many fields, including government, business, and science [14–16]. It uses a range from advanced online search engines like Google Search to self-driving cars created by startups like Waymo. To put it differently, AI has permeated almost every facet of contemporary society, ranging from communication and transportation to banking and medicine [17, 18].

Recent years have seen tremendous progress in the field of AI, especially with the advent of deep learning and transformer architecture [19]. A subtype of machine learning known as "deep learning" uses multi-layered neural networks to accomplish complicated tasks, while transformer

architecture is a cutting-edge method that enhances the visual representation of natural language processing in Figure 11.1. The early 2020s saw a boom in AI research and development as a result of these developments, and experts believe AI will continue to be crucial in determining how technology and society as a whole develop in the future.

The field of AI was pioneered by notable individuals like Alan Turing, whose studies and advancements date back to the mid-1950s. Since its official founding in 1956, AI has experienced multiple cycles of development and decrease, or "AI winters." Progress stalled during these times due to a decrease in funding, research, and interest in the topic. However, with the introduction of deep learning techniques that outperformed earlier AI approaches around 2012, the subject saw a rebirth in funding and attention.

In addition, the release of transformer architecture, a neural network model intended for natural language processing in 2017 contributed to the acceleration of AI research [20]. Significant advances in AI have been made during this time of increased attention and funding, including advances in machine learning, CV, and natural language processing. American businesses, academic institutions, and research facilities are currently at the forefront of these advancements, having made large expenditures in AI research and development [21]. Because of this, AI is developing quickly, and new use cases and applications are appearing in a variety of industries, including banking and healthcare.

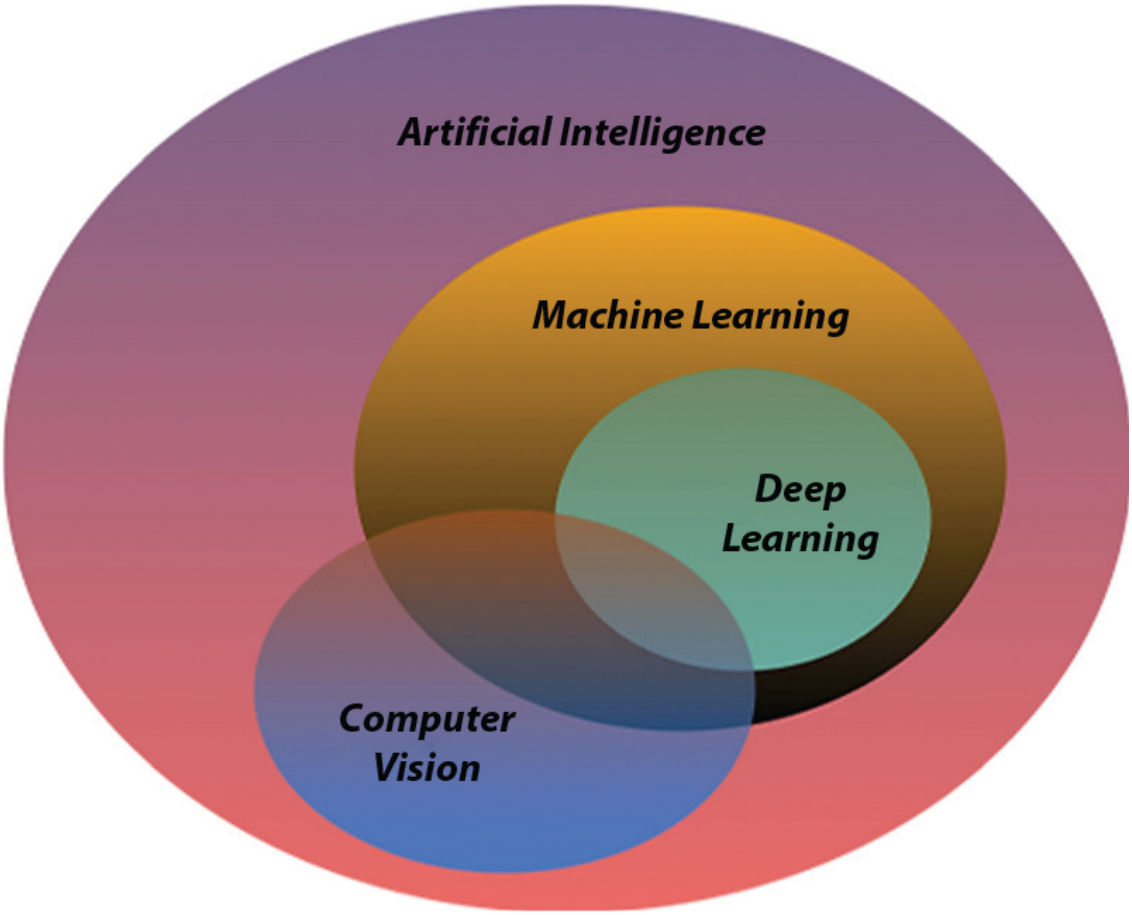


Figure 11.1 Venn diagram of artificial intelligence, machine learning, deep learning, and computer vision.

The assimilation of AI into various domains of society has led to a significant shift towards more automation, data-driven decision-making, and the widespread use of AI systems across multiple industries [22]. This development raises concerns about the long-term repercussions, ethical issues, and possible risks linked with AI and has important ramifications for labor markets, healthcare, government, industry, and education [23]. Regulations are becoming more and more necessary as AI develops to guarantee its safe and advantageous application.

AI research is a multidisciplinary field drawing upon a wide range of techniques and methodologies it is shown in Figure 11.2 [24]. Among these are statistical techniques, formal reasoning, artificial neural networks, search and optimization for mathematics, and so on. In order to better comprehend and mimic human intelligence, the field also integrates ideas from behavioral science, linguistics, philosophy, the field of neuroscience, and other academic fields [25, 26]. Advancements in AI have led to the development of intelligent machines that can perform tasks that were once thought to be exclusive to humans, such as natural

language processing, image recognition, and decision-making. However, this progress raises concerns about the impact of AI on employment, privacy, and security. Therefore, it is critical to balance the potential benefits of AI with the need to address ethical and societal issues associated with its use.

AI is a rapidly growing field that seeks to create intelligent systems that can perform a wide range of intellectual tasks that are typically associated with humans. To achieve this level of intelligence, AI researchers employ various tools and techniques such as machine learning algorithms, natural language processing, CV, and robotics, which enable the system to learn from data, understand and respond to human language, interpret visual data, and interact with the physical world [27].

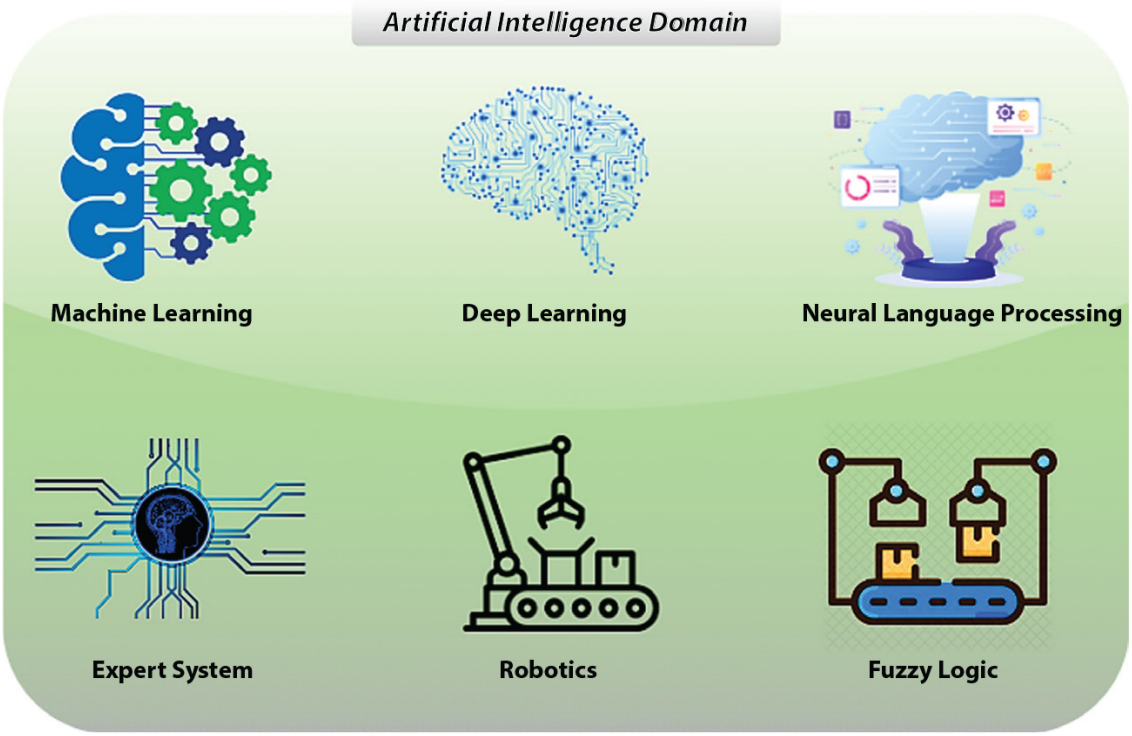


Figure 11.2 Various domains of artificial intelligence.

The ultimate goal of AI is to create systems that can exhibit general intelligence, which is the ability to perform intellectual tasks that are not limited to specific domains. This involves developing systems that can reason, understand, and process knowledge, plan and make decisions, learn from past experiences, and communicate using natural language [28].
Achieving this level of intelligence requires the integration of various AI techniques and tools to create systems that can perform a wide range of tasks with human-like intelligence and efficiency.

Moreover, the field of AI is not just about creating intelligent machines but also about understanding the nature of intelligence itself. AI researchers explore how humans think and learn and how these processes can be modeled and implemented in computer systems. The integration of AI with cognitive science, psychology, neuroscience, and other related fields has led to a deeper understanding of the mind and intelligence and has opened up new possibilities for developing AI systems.

AI is a rapidly evolving field that has the potential to revolutionize various industries by creating intelligent systems capable of performing a wide range of tasks with human-like intelligence and efficiency [29]. Achieving this level of intelligence requires the use of various tools and techniques, including deep learning, machine learning, and natural language processing, among others. It is crucial to address the ethical, societal, and economic implications of AI technology as it continues to advance and become increasingly sophisticated. The benefits of AI must be maximized while mitigating potential risks, and this requires interdisciplinary collaboration and a deep understanding of the nature of intelligence itself. AI researchers must work closely with experts in fields such as ethics, law, psychology, and sociology to ensure that AI is developed and deployed in a way that benefits society as a whole.

Furthermore, the interdisciplinary nature of AI research is key to its ongoing development and the realization of its full potential. Drawing upon insights from a wide range of fields, AI researchers can identify new and innovative ways to create intelligent systems that can transform various industries and improve people's lives. We must continue to invest in AI research and development, as it has the potential to revolutionize various aspects of society and bring about significant positive change.

11.2.1 Some Important Techniques

In this discussion, we will cover some crucial techniques that are commonly used in the field of AI. These techniques are instrumental in enabling machines to simulate human cognitive abilities such as learning, reasoning, and problem solving, as shown in Figure 11.3. We will explore each technique in detail:

a. Classifiers and Statistical Learning Method:

AI applications can be classified into two broad categories, namely, classifiers and controllers. Classifiers are methods that find the closest match by using pattern matching. Supervised learning, in which the algorithm is trained using labeled data, allows them to be adjusted depending on selected samples. A dataset is created when all of the observations are joined with their class labels. Each pattern or event is assigned a specific predetermined class. A fresh observation is categorized once it is obtained by using prior knowledge.

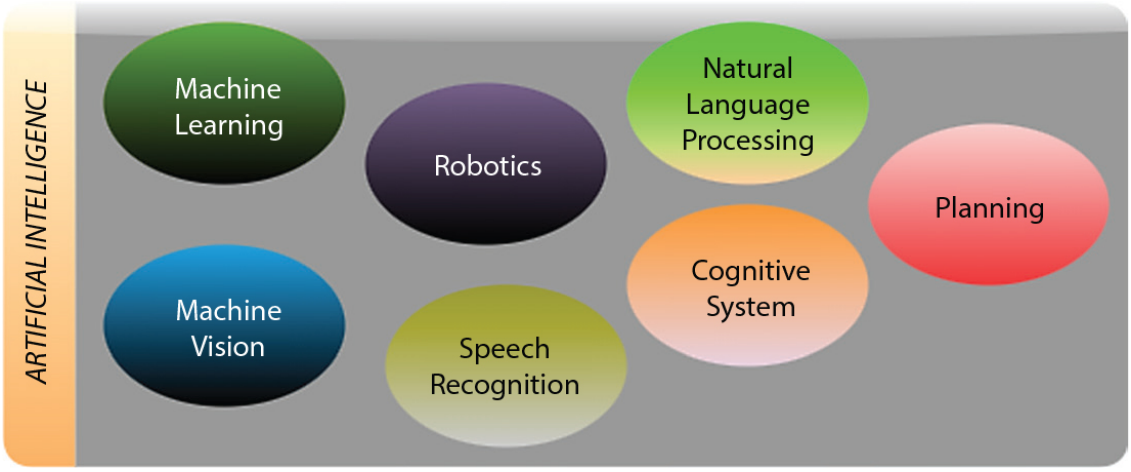


Figure 11.3 Features of artificial intelligence.

There are many kinds of classifiers in use, and one of the simplest and most widely used symbolic machine-learning algorithms is the decision tree. Another widely used algorithm until the mid-1990s was the K-nearest neighbor algorithm [30, 31], and conversely, kernel techniques like support vector machines (SVM) [32–34] moved out of the 1990s' k-nearest neighbor. Because of its scalability, a Bayes classifier that is naive is purportedly the most popular learner at Google. Classifiers that use neural networks are also employed.

This set of AI applications matches patterns to find the closest match. They can be improved via supervised learning based on selected instances after being trained on tagged data. There are various kinds of classifiers in use, such as decision trees, K-nearest neighbor algorithms, kernel methods, naive Bayes classifiers, and neural networks.

b. Search and Optimization:

AI is becoming a vital tool for resolving complicated issues by shrewdly sorting through a wide range of potential answers. AI uses two primary search strategies state space search and local search to achieve this.

State space search involves exploring all possible paths from the initial state to the goal state. This method is useful when there are a limited number of possible solutions, and the path to reach the goal is well-defined. State space search algorithms are typically used in problems such as route planning, where the goal is to find the shortest path between two points. This technique can also be applied to more complex problems such as game playing, where the goal is to find the best sequence of moves to win the game. State space search algorithms can be computationally expensive, but they guarantee that the optimal solution will be found if it exists.

On the other hand, the local search focuses on finding a solution by making small incremental changes to the current state until an acceptable solution is found. This method is useful when there are a large number of possible solutions, and it is difficult to determine the optimal path to reach the goal. Local search algorithms are commonly used in optimization problems such as scheduling, where the goal is to find the best sequence of tasks to complete in a given time frame. This technique can also be applied to machine learning problems such as clustering, where the goal is to group similar data points. Local search algorithms are often faster than state space search algorithms, but they may not always find the optimal solution.

c. Deep Learning:

An AI method called “deep learning” makes use of multiple layers of coupled neurons to extract high-level properties from raw data inputs [35]. These layers are intended to gradually examine and interpret data, with each layer adding to the information obtained in the previous layer shown in Figure 11.4. In image processing, for example, the lowest layers of a deep learning network may be in charge of recognizing fundamental properties such as edges and shapes. As the data progresses through the network, higher layers may recognize more complex patterns and concepts, such as letters, digits, and even faces. This hierarchical technique

enables deep learning algorithms to learn and recognize detailed patterns and correlations in data, making them appropriate for several applications in domains including CV, natural language processing, and speech recognition [36].

An important development in AI is deep learning. It has greatly enhanced the functionality of several systems, such as filtering images, speech recognition, natural language processing, CV, and many more [37, 38]. The explanation behind deep learning's outstanding performance across numerous areas remains unknown as of 2023. However, it is widely acknowledged that two things contributed significantly to its unexpected success between 2012 and 2015 [39].

- First and foremost, the extraordinary development in computer power has been critical to deep learning's success. Switching to GPUs resulted in a hundredfold improvement in speed, which was useful for processing larger and more complicated datasets. This allows deep learning algorithms to handle increasingly sophisticated neural networks and execute more difficult calculations in real time.
- Second, the availability of large volumes of training data has benefited deep learning. The massive curated datasets used for benchmark testing have allowed deep learning algorithms to learn and generalize more effectively. The more data the algorithms can train on, the better their ability to anticipate future outcomes. These data enable deep-learning computers to learn from a variety of sources and uncover previously unknown patterns.

In general, deep learning's success can be due to a combination of two factors: increased computer power and the availability of massive amounts of training data. With the growth of technology, we may expect deep learning to evolve and make even greater advances in the field of AI.

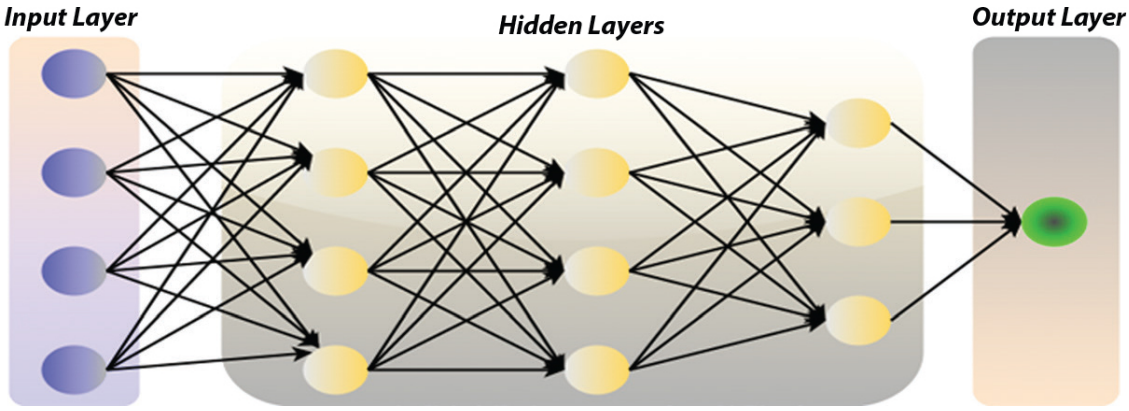


Figure 11.4 Deep learning architecture.

d. Artificial Neural Networks:

Artificial neural networks are machine learning models based on the structure of the human brain [40, 41]. Artificial neurons, also known as nodes, serve as the foundation of a neural network. These nodes are intended to represent the neurons in a biological brain [42].

The goal of an artificial neural network is to recognize patterns in data. Once taught, it can recognize the patterns in new data. A neural network has three major components: an input layer, one or more hidden layers, and an output layer.

Every node in a neural network performs a function on the data that it receives. As the neural network learns, the weights of the connections between nodes are modified to help the network produce more accurate predictions. When the weighted sum of a node's inputs reaches the defined threshold, the node's output is sent to the next layer [43].

A neural network is often referred to as a deep neural network if it contains at least two hidden layers. The number of layers in a neural network determines how complicated the patterns it can identify [44]. Deep neural networks are especially effective for tasks such as image identification and natural language processing.

During the training phase, neural networks, a subset of machine learning algorithms, use local search to ascertain the proper weights for each input. The most popular method for training these networks is the backpropagation algorithm. Neural networks can recognize patterns in big

datasets and predict complicated relationships between inputs and outputs by utilizing a sophisticated network of interconnected nodes. Neural networks are extremely adaptable tools with a wide range of applications because they can theoretically learn any function [43].

11.2.2 Types of Artificial Intelligence

If we are asked to describe the many types of AI systems, we should categorize them according to the functions that they have [45]. Rule-based systems, which follow a predetermined set of rules—machine learning-based systems, which learn from data without explicit programming; and neural networks, which draw inspiration from the human brain and can be trained to identify patterns and anticipate outcomes—are the three main types of AI systems [46]. Knowing the differences between these AI systems might help you decide which to utilize in a given situation [47, 48]. Each of them has distinct characteristics and uses falls into the following categories based on how AI-based systems function, as shown in Figure 11.5.

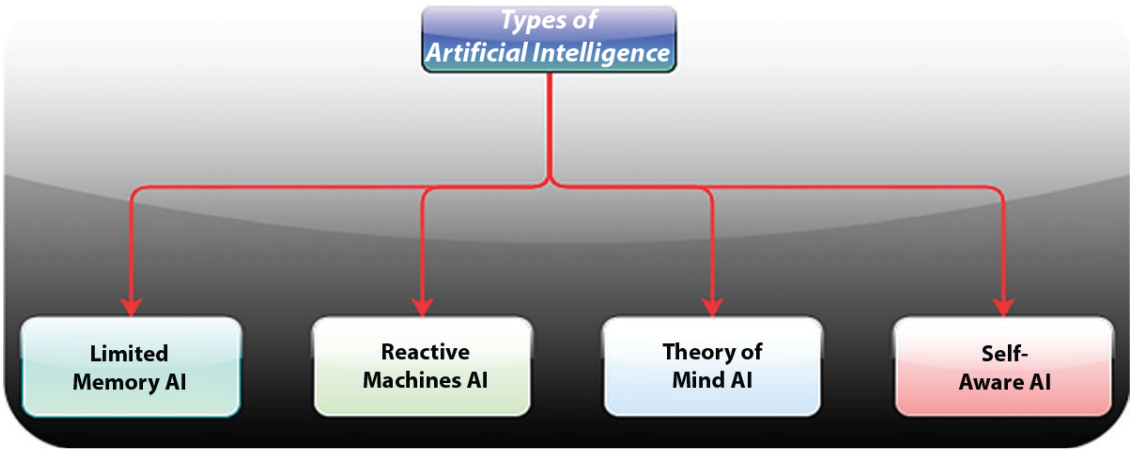


Figure 11.5 Types of artificial intelligence.

a. Limited Memory AI:

A system known as limited memory AI (MLAI) may analyze historical facts stored in its memory to make intelligent decisions. Limited memory AI stores previous experiences in its short-term or temporary memory and uses them to assess future actions, in contrast to other AI systems that only use real-time data to make decisions. This method allows limited memory AI to gradually enhance its decision-making capabilities. This

technique can be especially helpful in scenarios like financial forecasting or medical diagnostics when past data are crucial in identifying the best course of action.

In the automotive sector, self-driving automobiles represent a major technological leap. Because these cars are equipped with limited memory AI, they make snap decisions based on information gathered in the recent past. This aids in their decision-making while driving and helps them avoid any possible collisions. Multiple sensors are used by self-driving cars to detect their environment, including other vehicles on the road, traffic lights, hilly roads, and pedestrians crossing the street. Real-time data are gathered by the sensors, and the AI system uses that information to determine how the car should travel. The car learns to adapt to changes in its surroundings and make the safest driving decisions by repeating this procedure continuously while it is moving.

b. Reactive Machines AI:

AI that solely relies on current data is known as reactive AI. Reactive AI cannot observe its surroundings and draw conclusions that would direct its behavior in the future, in contrast to other AI systems. Rather, it only reacts to the current state of its surroundings and carries out a limited set of predetermined functions.

The IBM Chess software, which is well-known for defeating world champion Garry Kasparov in a sequence of games in 1997, is one notable example of reactive AI. Without any long-term planning or strategy, the program was made to examine the chess board and choose the optimal move only based on the pieces' current positions. Reactive AI can be quite useful in some situations where a focused concentration is suitable, such as playing games like chess, even though its flexibility and adaptability are limited.

c. Theory of Mind AI:

The Theory of Mind AI (AI) is a highly developed kind of intelligence that is thought to be the next development beyond machine learning. Because it is made with an emphasis on emotional intelligence, this type of com-

puter is deemed to be very important for psychology since it helps to better understand human beliefs and thinking. The goal of Theory of Mind AI development is to build machines that are more like humans in terms of their ability to reason, think, and make decisions. The ultimate objective is to build machines that can comprehend and forecast human behavior, feelings, and intentions. This would represent a huge advancement in AI.

However, creating the Theory of Mind AI is a difficult procedure that calls for in-depth study and development in addition to the fusion of numerous technologies. Even though this field has made great strides, a fully realized and useful Theory of Mind AI is still some ways off. However, this technology has a wide range of possible uses, and it might have a big impact on industries like education, healthcare, and psychology.

d. Self-Aware AI:

Curiosity and worry have been raised about the prospect of reaching a level of AI where robots possess awareness and become self-aware. Even while the concept of superintelligence in the future is still conceivable, this kind of AI is today unrealistic. Notable individuals like Elon Musk and the late Stephen Hawking have consistently alerted us to the possible dangers associated with the development of AI. They have stressed how crucial it is to create AI with safety and ethics as top priorities. It is critical to consider both the advantages and potential drawbacks of AI as we continue to explore its possibilities.

11.2.3 Role and Benefits of AI

AI has been transforming various domains and industries, providing innovative solutions and enhancing efficiency. From healthcare to finance, transportation education, AI has been changing the dimensions of work and paving the way for new opportunities. Its applications include natural language processing, CV, deep learning, and machine learning, among others. With its ability to analyze vast amounts of data, AI has enabled businesses to make better decisions, automate processes, and improve customer experiences. As technology continues to advance, we can expect

AI to play a more prominent role in shaping the future of work. Some are discussed below:

- ***Manufacturing:*** Demand forecasting, supply chain management, quality control, and scheduling precision can all be enhanced by AI. It can augment human capabilities and increase capital efficiency by co-operating with humans. This allows people to concentrate on activities that are most beneficial to their roles.
- ***Defense and Security:*** AI offers a wide range of real-world uses, such as risk terrain analysis, cyber defense, information gathering, and the identification of unusual human behavior. It can recognize and follow possible threats, react to cyberattacks, examine geographic data to find possible risk areas, and spot unusual human behavior to spot potential risks.
- ***Energy:*** Energy system modeling and forecasting help optimize power usage and balance by analyzing data from various sources to predict future energy demands and supply. AI and machine learning can refine these models, reduce energy costs, increase sustainability, and improve grid reliability.
- ***Financial Services:*** Automation can guarantee the early identification of financial risks and system malfunctions, hence mitigating malevolent intent in financial systems, including but not limited to market manipulation and abnormal trading.
- ***Law Enforcement:*** AI has emerged as a crucial instrument for law enforcement to ensure public safety. Authorities are now able to identify and apprehend offenders more quickly thanks to the use of predictive analytics, facial recognition, and speech recognition technologies. It has been demonstrated that these technologies are useful in deterring crime and safeguarding citizens.

- ***Science and Technology:*** AI holds immense potential in aiding scientists and researchers in their explorations by utilizing robotic systems to test theories. This approach offers new opportunities for faster, more accurate, and cost-effective experimentation, paving the way for scientific breakthroughs to be made more accessible than ever before.

11.3 Computer Vision

CV [49, 50] is a rapidly evolving area of study, within the realm of AI [51]. It focuses on creating techniques and algorithms that allow machines to interpret and understand information in how humans do. Essentially CV aims to teach machines how to perceive and make sense of data like images and videos. This involves processes such as image manipulation recognizing objects, detecting patterns, and more. By harnessing the capabilities of CV, we can enable machines to carry out an array of tasks with efficiency and precision. For instance, CV can help autonomous vehicles navigate roads safely by identifying traffic signals, pedestrians, and other cars. It also finds applications in fields for more accurate diagnoses by analyzing patterns in medical imagery [52]. The potential uses of CV are extensive promising to transform our interactions, with technology and the world at large.

CV is an evolving and fascinating field that merges computer science with intelligence to empower machines to analyze and understand visual data [53]. Through the utilization of algorithms and methods, CV equips machines, with the ability to comprehend, interpret, and derive insights from images and videos [54]. This technology finds applications across industries such as autonomous vehicles, healthcare, security, entertainment, and more. For example, it can be utilized for object detection and identification facial recognition, motion tracking well, and emotion detection among other functions. With progress in machine learning and image processing technologies, the potential impact of CV is vast. It is poised to revolutionize human-machine interactions, in the foreseeable future discussed in Figure 11.6. The capabilities of CV are boundless holding promise to transform domains by enabling machines to accomplish tasks that were previously deemed unattainable for them [55].

11.3.1 Application of Computer Vision

CV has many uses in many industries, including business, entertainment, healthcare, transportation, and daily living [56]. Some use cases are listed below:

- ***Healthcare:*** Leading the way in the application of CV technologies is the healthcare sector. Medical image analysis, which creates representations of organs and tissues to help medical practitioners make prompt and accurate diagnoses, is one of its important applications. Patients then have better treatment outcomes and longer life expectancies as a result. For example:
  - Analyzing skin lesions
  - Analyzing moles to detect tumors
  - Automatic X-ray interpreting
  - MRI scan-based symptom identification
- ***Agriculture:*** Through intelligent automation, CV applications improve the overall performance of the agricultural industry by increasing productivity and decreasing expenses. Both UAV and satellite imagery are useful for analyzing large areas of land and enhancing agricultural techniques. Applications for CV automate processes such as crop disease detection, soil moisture monitoring, weather forecasting, and crop yield prediction. Another important component of smart farming is CV-based animal monitoring.
- ***Autonomous vehicles:*** Through the analysis of real-time data from several cameras, autonomous vehicles employ CV to recognize objects such as pedestrians, road users, signs, and obstructions. They are also capable of making 3D maps.

  Machine learning (ML) and CV are used by semiautonomous cars to track the driver's actions. This technology uses the driver's head position, eye movement, and upper body motion to identify symptoms of exhaustion, drowsiness, and distraction. It lowers the likelihood of a driving incident by informing the driver when it notices specific warning flags.
- ***Security and safety:*** CV technology is used by both corporations and governments to improve the security of their sites, assets, and facilities. This entails keeping an eye on public areas, industrial locations, and high-security settings using cameras and sensors. Automatic warnings are triggered if any unexpected activity is identified, such as unlawful access to restricted locations.

  Furthermore, the use of CV technologies can improve personal safety at work and at home. For example, recognition technology can be used to track a variety of safety-related matters, such as identifying guests

or items delivered through front-door cameras, or detecting pets in real-time streams. This technology can be used in the workplace to track if employees are wearing the proper PPE and to provide reports or alerts when there is non-compliance.

- ***Retail*:** Retail chores like maintaining inventory levels and evaluating consumer behavior can be automated with CV. Retailers can improve product placement and store layouts, learn more about client preferences, and tailor the shopping experience. Additionally, CV makes it possible for novel shopping experiences like Amazon Go, which allows users to purchase more quickly and conveniently by avoiding the checkout procedure and using automatic payment.
- ***Banking:*** The banking industry has a lot of promise for CV technologies. Numerous procedures, including fraud detection, document verification, and customer service, can be automated with its assistance. CV algorithms can swiftly identify fraudulent activities and notify bank officials by examining thousands of papers in real time. This aids in avoiding monetary losses brought on by dishonest behavior. Furthermore, by examining consumer behavior and preferences, CV may also be utilized to improve the customer service experience. Banks can enhance customer happiness and loyalty by providing tailored advice to their clientele through the analysis of consumer data. In general, CV can benefit the banking industry by lowering expenses, increasing productivity, and enhancing customer care.
- ***Industrial:*** In recent years, CV technology has grown in popularity within the industrial sector. Osprey Informatics is one such program that is used to keep an eye on the condition of vital infrastructure. Since this technology allows for real-time video footage analysis and monitoring, it is very helpful for industrial facilities, remote wells, job activity, and site security. Businesses can swiftly spot any issues and take preventative action to stop them from getting worse by utilizing CV. This enhances operations and lowers downtime in addition to enhancing safety and security. All things considered, CV is shown to be a useful instrument for raising production and efficiency in the industrial sector.

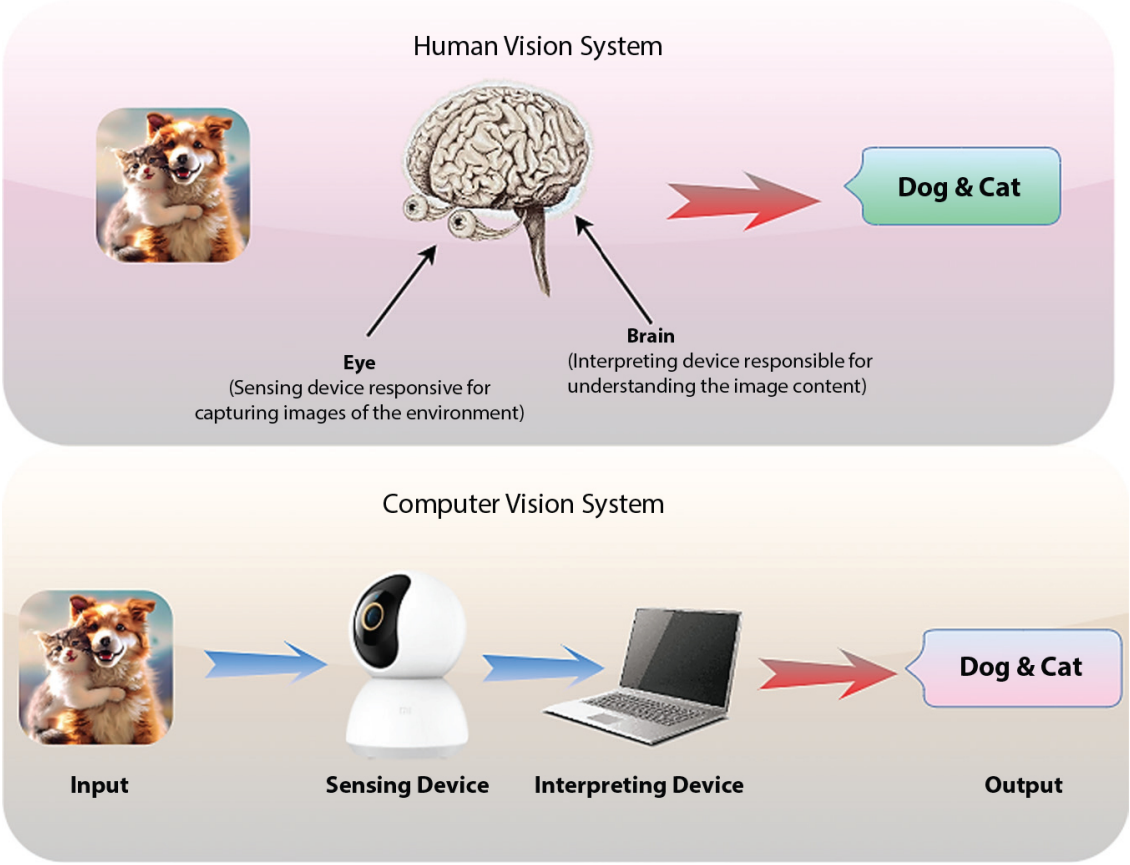


Figure 11.6 Flow of human vision and computer vision.

Applications such as object classification, identification, detection, segmentation, verification, recognition, and landmark detection depend heavily on CV techniques. These methods facilitate object recognition in digital photos and high-dimensional data extraction to provide numerical or symbolic real-world information [57].

CV is a field of study that involves teaching machines to recognize and interpret visual data. This technology has many advantages, including increased accuracy, efficiency, and the ability to process large amounts of data quickly. However, creating these algorithms requires a lot of training databases, and interpreting visual data can be challenging. Additionally, automating functions that were once performed by humans raises ethical questions that must be addressed [58]. Despite these challenges, CV is a rapidly evolving field that has the potential to revolutionize many industries and improve our lives in countless ways.

11.3.2 Working of Computer Vision

CV technology aims to mimic the functionality of the human visual system by using cameras to capture images or videos [54, 59]. These visual

inputs are then processed by advanced algorithms to extract meaningful data that can be utilized for various applications. The CV process involves several critical steps, which are illustrated in Figure 11.7 [60]. These steps include image acquisition, image preprocessing, feature extraction, object detection, and image classification. Each of these steps plays a vital role in the overall process, and the accuracy of the output depends on the effectiveness of each step [61].

- ***Image Acquisition:*** To begin with, the initial step in capturing visual information is to use a camera to take a picture or record a video. In essence, the process is similar to how our eyes function in gathering and interpreting visual data.
- ***Image Processing:*** After being taken, the image is processed into a digital representation, usually a pixel matrix. The values for the red, green, and blue (RGB) color channels are contained in each pixel in this matrix, which represents a very small piece of the image. The intensity of each color in that pixel is indicated by these numbers, which range from 0 to 255.
- ***Feature Extraction:*** The process of extracting features from the image that are pertinent to the current task is the following phase. This can entail locating the image's corners, edges, or textures. With feature extraction, the most crucial information in an image is preserved while the image's complexity is decreased.
- ***Pattern Recognition:*** The process of extracting features from the image that are pertinent to the current task is the following phase. This can entail locating the image's corners, edges, or textures. With feature extraction, the most crucial information in an image is preserved while the image's complexity is decreased.
- ***Decision Making:*** The final stage is to use the recognized patterns to make judgments or predictions. For example, in a facial recognition system, the computer may compare the attributes of a taken image to a database of known faces to identify the individual.

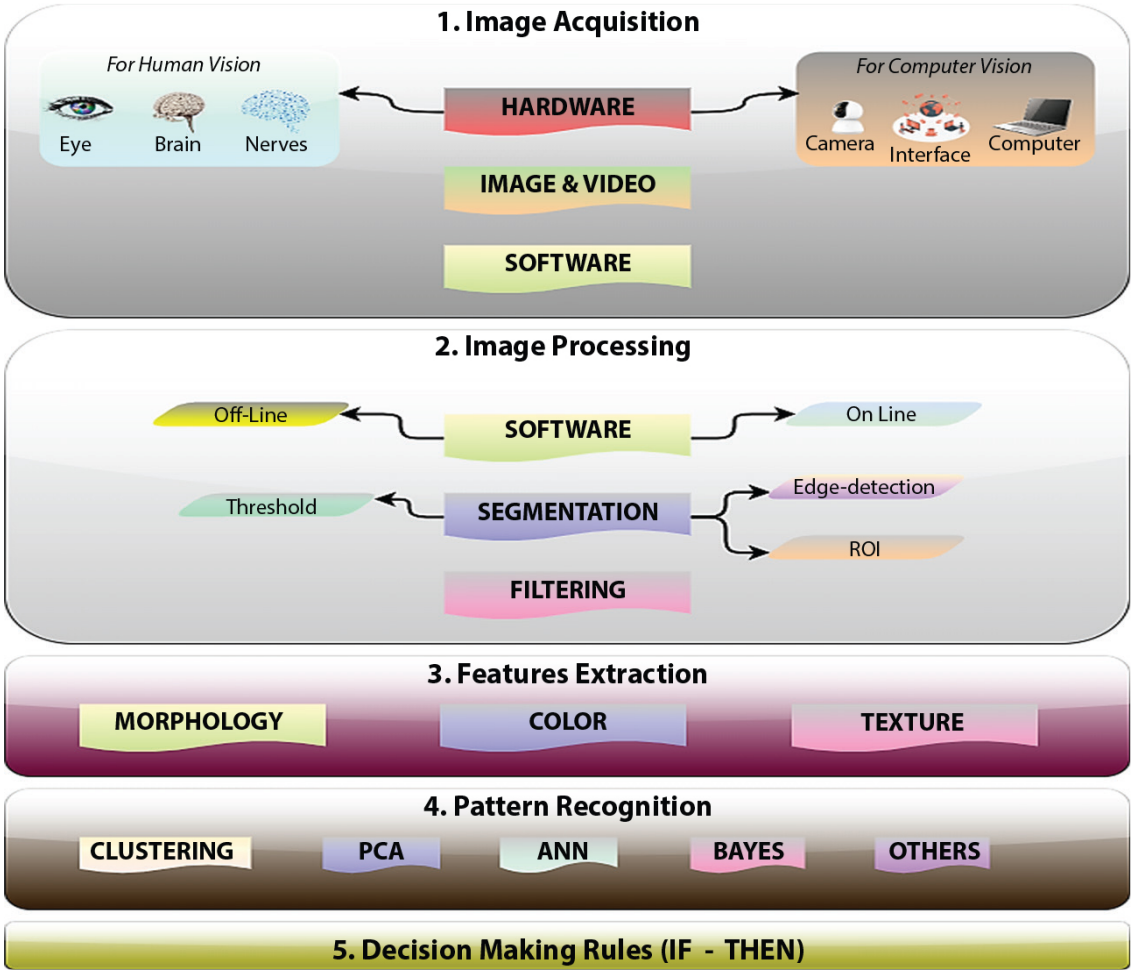


Figure 11.7 Working flow of computer vision.

CV is an advanced procedure that enables computers to perceive and comprehend the visual world. It allows machines to interpret and analyze digital images from the real world, providing a wide range of applications [62]. With CV, we can develop a variety of innovative technological solutions such as medical imaging, facial recognition, and driverless cars. This technology has the potential to revolutionize the way we interact with the world around us and make our lives easier and more efficient.

11.3.3 Importance of Computer Vision for IAS

CV is the technology that allows digital images and movies to be interpreted and understood by machines [63]. Because it greatly increases the capabilities and efficiency of intelligent automation systems (IAS) [64] in a variety of industries, it has become an indispensable part of the systems. Machines that were previously incapable of performing complicated tasks can now recognize, analyze, and classify visual input with the aid of CV. CV, for example, can be used in manufacturing to check items for

flaws, keep an eye on the production line, and spot possible safety risks. CV can be used in healthcare to help with surgery, monitor patients, and identify illnesses. All things considered, CV has completely changed how robots interact with the outside environment, giving them increased intelligence, accuracy, and dependability [65].

- ***Invoice Processing:*** Automating the processing of bills, which formerly required manual data entry, is possible with CV technology. This can be done by using photos of invoices to identify and extract data, including order numbers, vendor names, and remittance information. Through the use of this technology, payments can be processed more quickly and accurately by streamlining the payment process.
- ***Know-Your-Customer (KYC) Automated Onboarding:*** KYC compliance is essential in the financial sector. Financial firms can onboard clients remotely by using CV-enabled remote video-based customer identification procedures. This lowers the possibility of fraud while also expediting the onboarding process.
- ***Insurance Claims Automation*****:** The process of filing an insurance claim can be expedited by using a CV. Clients can email images of their damaged goods and IAS (insurance adjusting services) will expeditiously assess the cost of repairs, authorize claims, and handle payment processing. By doing away with the requirement for manual approvals and inspections, this speeds up and improves the efficiency of the entire process.
- ***Equipment Safety Inspections*****:** CV can be used in manufacturing and other industries to monitor machinery in real time and spot any problems before they become serious ones. By approaching maintenance proactively, downtime can be avoided and operational effectiveness can be increased.
- ***Automated Retail Checkouts*****:** Retail establishments can completely transform their checkout procedure by incorporating CV technologies. Thanks to this cutting-edge technology, transactions can be automated, resulting in quick processing times and a flawless online shopping experience for clients. Upgrade to CV technology immediately to improve your customers' shopping experience and avoid having your retail business hampered by laborious and manual checkout processes.

- ***Medical Diagnostics*:** A crucial instrument in the medical field, the CV analyzes medical images efficiently to enable prompt and precise disease diagnosis as well as ongoing patient health monitoring. Its use has been shown to dramatically increase diagnosis timeliness and accuracy, making it an essential part of contemporary healthcare systems.
- ***Surveillance and Biometrics*:** Surveillance systems require CV because it allow automated environment monitoring to identify suspicious activity or unlawful entry. It is also essential to biometrics, helping to make safe processes of authentication and identification possible.

CV is essential to IAS because, apart from AI and machine learning, it can evaluate unstructured and semi-structured data. With the use of this technology, computers are now able to learn from data, improve over time, and automate labor-intensive, complex tasks that were previously laborious and prone to human error. As an indication of the expanding importance and adoption of this technology in IAS, the size of the global CV market was estimated to reach $10.56 billion in 2019 and is expected to climb [3, 66].

By enabling computers to “see” and comprehend their environment, CV is also revolutionizing other industries, as evidenced by the development of self-driving cars and the integration of AI into machine vision. Operations get increasingly independent, safer, and more efficient as a result of this transition.

11.4 Conclusion

IAS combine CV and AI to improve productivity, efficiency, and decision-making, thereby revolutionizing job and process automation. AI gives machines the ability to think like humans, evaluate data, identify patterns, and make decisions. This allows businesses to perform complicated jobs effectively without requiring continual human intervention. This dynamic integration offers a more intelligent and flexible approach than traditional automation, with AI algorithms that can learn from data and adjust to changing conditions.

By enabling robots to perceive visual stimuli, CV improves IAS and is necessary for tasks like object recognition, autonomous navigation, and quality assurance. In contexts where decision-making relies heavily on visual cues, this integration makes automation more precise and effective. Together, AI, CV, ML, and DL enhance automated processes and promote human-machine interaction, enabling decision-making and managing activities that are beyond the capabilities of humans. Automation systems will become smarter, more flexible, and more effective as these technologies advance and promise to completely change a number of industries.

11.5 Future Scope

Future developments in IAS utilizing CV and AI promise revolutionary breakthroughs in a range of industries. Improved autonomous systems will navigate and make judgments with more safety and precision, such as drones and self-driving cars. Healthcare will gain from better diagnoses, individualized therapies, and robotic surgery, while manufacturing will benefit from smarter factories with precise robots and streamlined processes thanks to CV and AI.

Advanced behavior analysis and facial recognition technologies in security and surveillance will improve public safety. These technologies will be used in agriculture and environmental monitoring to improve data analysis and implement sustainable practices. As AI and CV systems enhance human capacities, human-machine collaboration will improve, and material science and bioinformatics research and development will pick up speed.

Future IAS will be adaptable and scalable, always learning and changing with little assistance from humans. This will result in automation systems that are more intelligent and versatile, promoting efficiency and creativity in all industries.

References

1. Ertel, W., *Introduction to artificial intelligence*, Springer, 2018.

2. Minh, D., Wang, H.X., Li, Y.F., Nguyen, T.N., Explainable artificial intelligence: a comprehensive review. *Artif. Intell. Rev.*, 55, 1–66, 2022.
3. Dudukalov, E., Munister, V., Zolkin, A., Losev, A., Knishov, A., The use of artificial intelligence and information technology for measurements in mechanical engineering and in process automation systems in Industry 4.0. *J. Phys.: Conf. Ser.*, 1889, 052011, 2021, IOP Publishing.
4. Zeynal, H., Eidiani, M., Yazdanpanah, D., Intelligent substation automation systems for robust operation of smart grids. *2014 IEEE Innovative Smart Grid Technologies-Asia (ISGT ASIA)*, pp. 786–790, 2014.
5. Sarker, I.H., AI-based modeling: techniques, applications and research issues towards automation, intelligent and smart systems. *SN Comput. Sci.*, **3**, 2, 158, 2022.
6. Hanna, A., Bengtsson, K., Dahl, M., Erős, E., Götvall, P.-L., Ekström, M., Industrial challenges when planning and preparing collaborative and intelligent automation systems for final assembly stations, in: *2019 24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, IEEE, 2019.
7. Madakam, S., Ramaswamy, R., Tripathi, S., Internet of Things (IoT): A literature review. *J. Comput. Commun.*, **3**, 5, 164–173, 2015.
8. Adadi, A. and Berrada, M., Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). *IEEE Access*, **6**, 52138–52160, 2018.
9. Das, A. and Rad, P., Opportunities and challenges in explainable artificial intelligence (xai): A survey. 2020, arXiv preprint arXiv:2006.11371.
10. Vilone, G. and Longo, L., Explainable artificial intelligence: a systematic review. 2020, arXiv preprint arXiv:2006.00093.
11. Angelov, P.P., Soares, E.A., Jiang, R., Arnold, N.I., Atkinson, P.M., Explainable artificial intelligence: an analytical review. *Wiley Interdiscip. Rev.: Data Min. Knowl. Discov.*, **11**, 5, e1424, 2021.
12. Hunt, E.B., *Artificial intelligence*, Academic Press, 2014.
13. Ahmed, I., Jeon, G., Piccialli, F., From artificial intelligence to explainable artificial intelligence in industry 4.0: a survey on what, how, and where. *IEEE Trans. Ind. Inf.*, **18**, 8, 5031–5042, 2022.
14. Holzinger, A., Langs, G., Denk, H., Zatloukal, K., Müller, H., Causability and explainability of artificial intelligence in medicine. *Wiley Interdiscip. Rev.: Data Min. Knowl. Discov.*, **9**, 4, e1312, 2019.

15. Došilović, F.K., Brčić, M., Hlupić, N., Explainable artificial intelligence: A survey, in: *2018 41st International convention on information and communication technology, electronics and microelectronics (MIPRO)*, IEEE, 2018.
16. Holmes, J., Sacchi, L., Bellazzi, R., Artificial intelligence in medicine. *Ann. R. Coll. Surg. Engl.*, **86**, 334–8, 2004.
17. Samek, W. and Müller, K.-R., Towards explainable artificial intelligence, in: *Explainable AI: interpreting, explaining and visualizing deep learning*, pp. 5–22, 2019.
18. Gunning, D., Stefik, M., Choi, J., Miller, T., Stumpf, S., Yang, G.-Z., XAI—Explainable artificial intelligence. *Sci. Rob.*, **4**, 37, eaay7120, 2019.
19. Gunning, D. and Aha, D., DARPA's explainable artificial intelligence (XAI) program. *AI Mag.*, **40**, 2, 44–58, 2019.
20. Arrieta, A.B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf. Fusion*, **58**, 82–115, 2020.
21. Core, M.G., Lane, H.C., Van Lent, M., Gomboc, D., Solomon, S., Rosenberg, M., Building explainable artificial intelligence systems, in: *AAAI*, 2006.
22. Phillips, P.J., Phillips, P.J., Hahn, C.A., Fontana, P.C., Yates, A.N., Greene, K., Broniatowski, D.A., Przybocki, M.A., Four principles of explainable artificial intelligence, 2021.
23. Jiménez-Luna, J., Grisoni, F., Schneider, G., Drug discovery with explainable artificial intelligence. *Nat. Mach. Intell.*, **2**, 10, 573–584, 2020.
24. Islam, S.R., Eberle, W., Ghafoor, S.K., Ahmed, M., Explainable artificial intelligence approaches: A survey. 2021, arXiv preprint arXiv:2101.09429.
25. Ben Ayed, R. and Hanana, M., Artificial intelligence to improve the food and agriculture sector. *J. Food Qual.*, **2021**, 1–7, 2021.
26. Manickam, P., Mariappan, S.A., Murugesan, S.M., Hansda, S., Kaushik, A., Shinde, R., Thipperudraswamy, S., Artificial intelligence (AI) and internet of medical things (IoMT) assisted biomedical systems for intelligent healthcare. *Biosensors*, **12**, 8, 562, 2022.
27. Tang, A., Li, K.K., Kwok, K.O., Cao, L., Luong, S., Tam, W., The importance of transparency: Declaring the use of generative artificial intelli-

gence (AI) in academic writing. *J. Nurs. Scholarsh.*, **56**, 2, 314–318, 2024.

28. Huang, M.-H. and Rust, R.T., Artificial intelligence in service. *J. Serv. Res.*, **21**, 2, 155–172, 2018.
29. Vinuesa, R., Azizpour, H., Leite, I., Balaam, M., Dignum, V., Domisch, S., Felländer, A., Langhans, S.D., Tegmark, M., Fuso Nerini, F., The role of artificial intelligence in achieving the Sustainable Development Goals. *Nat. Commun.*, **11**, 1, 1–10, 2020.
30. Cunningham, P. and Delany, S.J., K-nearest neighbour classifiers-a tutorial. *ACM Comput. Surv. (CSUR)*, **54**, 6, 1–25, 2021.
31. Triguero, I., Maillo, J., Luengo, J., García, S., Herrera, F., From big data to smart data with the k-nearest neighbours algorithm, in: *2016 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData)*, IEEE, 2016.
32. Jakkula, V., Tutorial on support vector machine (svm). *Sch. EECS Wash. State Univ.*, **37**, 2.5, 3, 2006.
33. Schuldt, C., Laptev, I., Caputo, B., Recognizing human actions: a local SVM approach, in: *Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004*, IEEE, 2004.
34. Vishwanathan, S. and Murty, M.N., SSVM: a simple SVM algorithm, in: *Proceedings of the 2002 International Joint Conference on Neural Networks. IJCNN'02 (Cat. No. 02CH37290)*, IEEE, 2002.
35. Goodfellow, I., Bengio, Y., Courville, A., *Deep learning*, MIT press, 2016.
36. Yang, Y., Ye, Z., Su, Y., Zhao, Q., Li, X., Ouyang, D., Deep learning for. *Acta Pharm. Sin. B*, **9**, 1, 177–185, 2019.
37. LeCun, Y., Bengio, Y., Hinton, G., Deep learning. *Nature*, **521**, 7553, 436–444, 2015.
38. Kamilaris, A. and Prenafeta-Boldú, F.X., Deep learning in agriculture: A survey. *Comput. Electron. Agric.*, **147**, 70–90, 2018.
39. Buduma, N., Buduma, N., Papa, J., *Fundamentals of deep learning*, O'Reilly Media, Inc., 2022.
40. Abdolrasol, M.G., Hussain, S.S., Ustun, T.S., Sarker, M.R., Hannan, M.A., Mohamed, R., Ali, J.A., Mekhilef, S., Milad, A., Artificial neural networks based optimization techniques: A review. *Electronics*, **10**, 21, 2689, 2021.

41. Zou, J., Han, Y., So, S.-S., Overview of artificial neural networks, in: *Artificial neural networks: methods and applications*, pp. 14–22, 2009.
42. Yegnanarayana, B., *Artificial neural networks*, PHI Learning Pvt. Ltd, 2009.
43. Basheer, I.A. and Hajmeer, M., Artificial neural networks: fundamentals, computing, design, and application. *J. Microbiol. Methods*, **43**, 1, 3–31, 2000.
44. Graupe, D., *Principles of artificial neural networks*, vol. 7, World Scientific, 2013.
45. Strelkova, O., *Three types of artificial intelligence*, Intelligent Information Management, **15**, 4, July 24, 2023.
46. Martinez, R., Artificial intelligence: Distinguishing between types & definitions. *Nev. Law J.*, **19**, 3, 9, 2019.
47. Martinez, R., Artificial intelligence: Distinguishing between types & definitions. *Nev. Law J.*, **19**, 1015, 2018.
48. Weiss, G., *Multiagent systems: a modern approach to distributed artificial intelligence*, MIT press, 1999.
49. Voulodimos, A., Doulamis, N., Doulamis, A., Protopapadakis, E., Deep learning for computer vision: A brief review. *Comput. Intell. Neurosci.*, **2018**, 1, 7068349, 2018.
50. Stockman, G. and Shapiro, L.G., *Computer vision*, Prentice Hall PTR, 2001.
51. Jähne, B., Haussecker, H., Geissler, P., *Handbook of computer vision and applications*, vol. 2, Citeseer, 1999.
52. Granlund, G.H. and Knutsson, H., *Signal processing for computer vision*, Springer Science & Business Media, 2013.
53. Weinstein, B.G., A computer vision for animal ecology. *J. Anim. Ecol.*, **87**, 3, 533–545, 2018.
54. Parker, J.R., *Algorithms for image processing and computer vision*, John Wiley & Sons, 2010.
55. Umbaugh, S.E., *Digital image processing and analysis: human and computer vision applications with CVIPtools*, CRC press, 2010.
56. Szeliski, R., *Computer vision: algorithms and applications*, Springer Nature, 2022.
57. Blehm, C., Vishnu, S., Khattak, A., Mitra, S., Yee, R.W., Computer vision syndrome: a review. *Surv. Ophthalmol.*, **50**, 3, 253–262, 2005.

58. Weickert, J., Applications of nonlinear diffusion in image processing and computer vision, *Deutsche Nationalbibliothek Reihe O*, 2000.
59. Meer, P., Robust techniques for computer vision, in: *Emerging topics in computer vision*, pp. 107–190, 2004.
60. Martynenko, A., Computer Vision for Real-Time Control in Drying. *Food Eng. Rev.*, **9**, 91–111, 2017.
61. Mery, D., Svec, E., Arias, M., Riffo, V., Saavedra, J.M., Banerjee, S., Modern computer vision techniques for x-ray testing in baggage inspection. *IEEE Trans. Syst. Man Cybern.: Syst.*, **47**, 4, 682–692, 2016.
62. O'Mahony, N., Campbell, S., Carvalho, A., Harapanahalli, S., Hernandez, G.V., Krpalkova, L., Riordan, D., Walsh, J., Deep learning vs. traditional computer vision, in: *Advances in Computer Vision: Proceedings of the 2019 Computer Vision Conference (CVC), Volume 1, 1*, Springer, 2020.
63. Taheri-Garavand, A., Fatahi, S., Omid, M., Makino, Y., Meat quality evaluation based on computer vision technique: A review. *Meat Sci.*, **156**, 183–195, 2019.
64. Tyagi, A.K., Fernandez, T.F., Mishra, S., Kumari, S., Intelligent automation systems at the core of industry 4.0, in: *International conference on intelligent systems design and applications*, Springer, 2020.
65. Gilabert, E. and Arnaiz, A., Intelligent automation systems for predictive maintenance: A case study. *Rob. Comput. Integr. Manuf.*, **22**, 5-6, 543–549, 2006.
66. Huysentruyt, K., Kjoersvik, O., Dobracki, P., Savage, E., Mishalov, E., Cherry, M., Leonard, E., Taylor, R., Patel, B., Abatemarco, D., Validating intelligent automation systems in pharmacovigilance: insights from good manufacturing practices. *Drug Saf.*, **44**, 261–272, 2021.

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